Polish Electricity Demand & Forecast Performance Analysis

An end-to-end data analytics project using Python, PostgreSQL, SQL and Power BI

https://github.com/bkluj/EnergyMarketBoard

Tools: Python, Pandas, PostgreSQL, SQL, Power BI

Description

This project analyzes electricity demand and forecast performance in the Polish power system.

The main objective was to identify recurring demand patterns, evaluate differences between actual and forecasted system load, and determine the periods in which forecast errors were the largest.

Data from multiple reporting periods was processed and validated in Python, stored in PostgreSQL, analyzed using SQL, and presented in an interactive Power BI dashboard.

The final dataset contains 13,436 observations recorded at 15-minute intervals between January and June 2026.

The analysis showed that average forecast error remained relatively low, with a MAPE of 2.70%. Electricity demand followed clear intraday and weekly patterns, while the largest average forecast errors occurred around midday.

KPI	Result
Average actual load	18,498 MW
Maximum actual load	26,787 MW
MAE	491 MW
RMSE	646.44 MW
MAPE	2.70%

Problem statement

Electricity demand changes throughout the day and week, creating recurring load patterns in the power system. Accurate demand forecasts are important for generation planning and maintaining the balance between electricity supply and demand.

Differences between forecasted and actual system load may indicate periods in which electricity demand is more difficult to predict.

The objective of this analysis was to examine historical electricity demand in the Polish power system, evaluate forecast performance and identify recurring periods of higher demand and larger forecast errors.

Analytical questions

- How closely does forecasted system load follow actual load?
- What are the main intraday electricity demand patterns?
- During which hours is average system load the highest?
- During which hours are forecast errors the largest?
- How does electricity demand differ between weekdays and weekends?
- How frequently is system load overforecasted or underforecasted?
- Which observations show the largest forecast deviations?

Dataset overview

- **13,436 observations**
- **20 final analytical columns**
- **15-minute intervals**
- **Date range: 24 January 2026 – 14 June 2026**
- Actual total load in MW
- Forecasted total load in MW
- Time and calendar variables
- Forecast-error variables

Data source: https://www.pse.pl/web/pse-eng/data/polish-power-system-operation/load-of-polish-power-system

Data quality validation

Chronological validation identified two irregularities in the source data:

- a one-week gap between two reporting periods,
- a one-hour irregularity related to the daylight-saving-time transition.

These cases were investigated separately and were not treated as ordinary missing records or duplicated timestamps.

The validation process included checking missing values, duplicated observations, duplicated timestamps and time differences between consecutive records.

Project workflow

Raw Excel files

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Python and Pandas preprocessing

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Cleaned and enriched dataset

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PostgreSQL database

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SQL analysis

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Power BI dashboard

Data cleaning and preparation

- Located and sorted all `.xlsx` files.
- Loaded individual files with Pandas.
- Concatenated them into one DataFrame.
- Cleaned newline characters and spaces from column names.
- Removed duplicate rows.
- Checked missing values and duplicated timestamps.
- Created a complete datetime field from date and time interval columns.
- Sorted records chronologically.
- Checked time differences between consecutive observations.
- Exported the processed dataset.

```
dfs = [
    pd.read_excel("../data/raw/" + file_name)
    for file_name in sorted(os.listdir("../data/raw"))
]

df = pd.concat(dfs, ignore_index=True)
df.drop_duplicates(inplace=True)
df = df.sort_values("Datetime").reset_index(drop=True)
```

Feature engineering

Calendar features

- Hour
- Weekday
- Weekday number
- Week number
- Month
- Year
- Weekend flag

Forecast error

$$Forecast\ Error = Actual\ Load - Forecasted\ Load$$

Interpretation:

- positive value → underforecast,
- negative value → overforecast.

Absolute error

$$Absolute\ Error = |\ Forecast\ Error\ |$$

Absolute percentage error

$$APE = \frac{|\ Actual - Forecast\ |}{Actual} \times 100\%$$

Squared error

$$Squared\ Error = (Actual - Forecast)^2$$

Forecast status

- Underforecast
- Overforecast
- Exact

SQL and database

The final dataset was loaded into PostgreSQL using SQLAlchemy. SQL was used to calculate aggregations, compare weekdays and weekends, identify peak-load periods and find the largest forecast errors.

Example query:

```
SELECT
    hour,
    ROUND(AVG(absolute_error_mw)::NUMERIC, 2)
        AS average_absolute_error_mw
FROM fact_power_system_load
GROUP BY hour
ORDER BY average_absolute_error_mw DESC;
```

Main results

Metric	Result
Average actual load	18,498 MW
Average forecasted load	18,711 MW
Maximum actual load	26,787 MW
Minimum actual load	11,211 MW
MAE	491 MW
RMSE	646.44 MW
MAPE	2.70%

Key findings

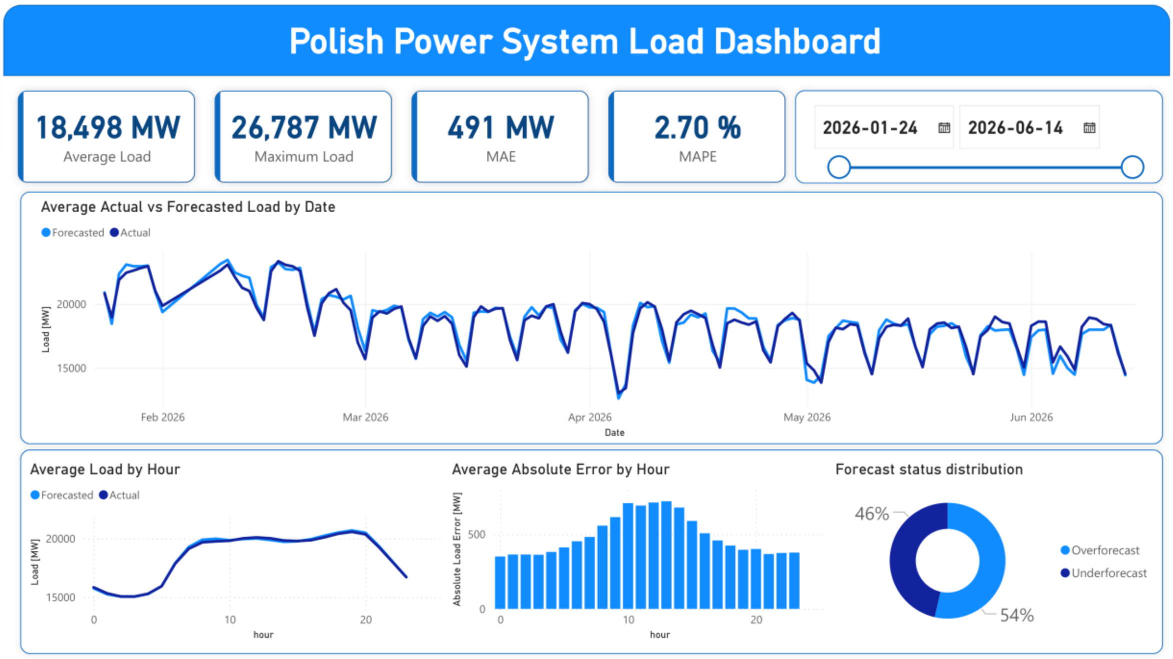
- The forecast showed relatively low average percentage error, with a MAPE below 3%
- Average forecasted load was slightly higher than average actual load, indicating a small overall overforecasting bias.
- 54.77% of observations were overforecasts.
- 45.23% were underforecasts.
- The highest average demand occurred around 19:00.
- The largest average forecast errors occurred around 12:00–13:00.
- Weekday load was around 19,511 MW.
- Weekend load was around 16,548 MW - so average weekend system load was approximately 15% lower than weekday load.

Analytical interpretation

The results indicate clear intraday and weekly electricity demand patterns. Average system load was significantly lower during weekends, while peak average demand occurred during evening hours. The largest forecast errors occurred around midday rather than during the evening demand peak, suggesting that forecast difficulty was not driven solely by the absolute level of system load.

Power BI dashboard

The interactive Power BI dashboard provides an overview of electricity demand and forecast performance. It allows users to analyze changes over time and compare forecast accuracy across different hours and reporting periods.



The dashboard includes KPI cards, a date slicer, actual versus forecasted load, average hourly demand, hourly MAE and forecast-status distribution.

Conclusions and next steps

The analysis showed that the forecast followed actual electricity demand relatively closely. However, forecast accuracy varied by hour, with the largest errors occurring around midday.

The project demonstrates practical skills in Python data preparation, PostgreSQL integration, SQL analysis and Power BI reporting.

Next steps

- integrate electricity price data,
- include renewable generation data,
- include cross-border electricity flows,
- analyze relationships between demand and electricity prices,
- incorporate weather and holiday variables,
- build a short-term load forecasting model,
- automate data ingestion and database updates.